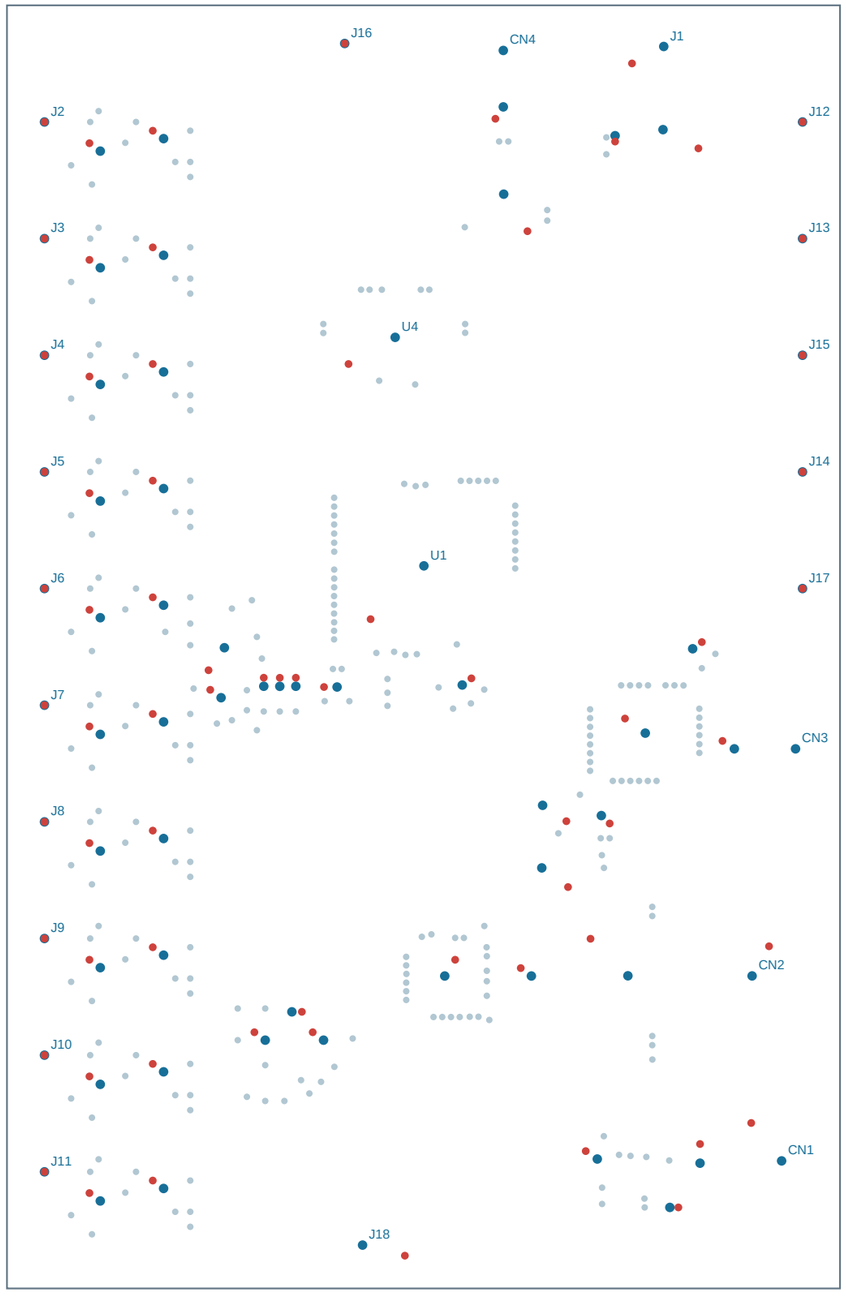
Assembly review - verified CAD geometry

412 parts per board | 333 top / 79 bottom | 70 directional parts reviewed | 59 CPL rotation updates

Top - top view

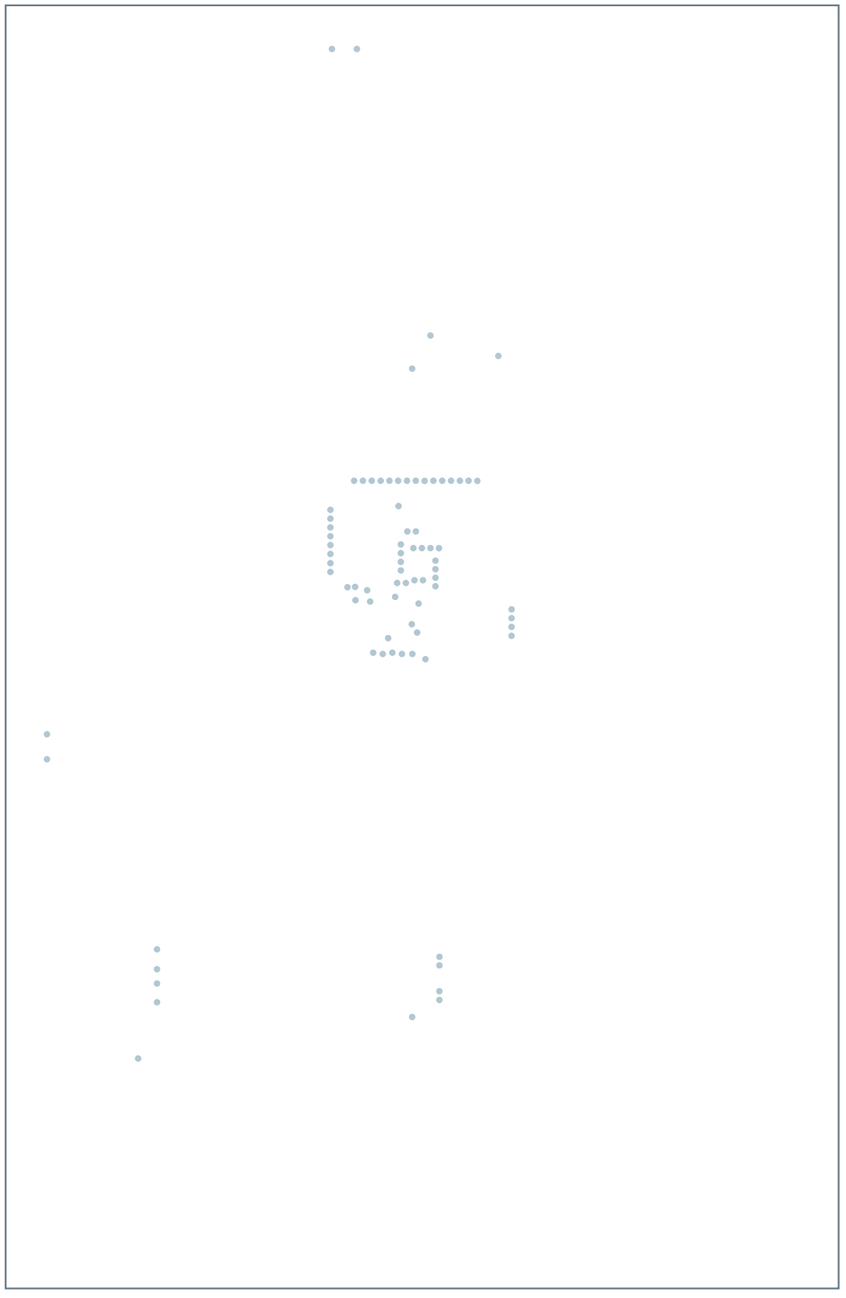


Red: pin 1 / A1 / cathode. Blue: reviewed centroid.

CPL XY uses top view on both sides; never mirror the CPL.

Numbered-pad detail pages define orientation.

Bottom - physical bottom view, X mirrored



Red: pin 1 / A1 / cathode. Blue: reviewed centroid.

CPL XY uses top view on both sides; never mirror the CPL.

Numbered-pad detail pages define orientation.

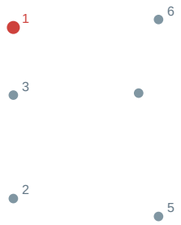
Numbered-pad direction review 1-12

All detail plots are top view, +X right / +Y up. Red marks pin 1, A1 or LED cathode; plots are not to scale.

CN1 | CPL 180 deg | Top

PJ-051BH

XY: 92.99, 15.31 mm



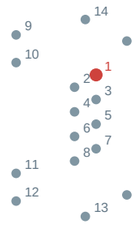
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

CN2 | CPL 0 deg | Top

LPJE101-0AHN

XY: 89.446, 37.51 mm



Exact purchased part / tape zero pending

manual insertion drawing checked Confirm exact connector outline, pin 1 and mating direction from drawing; hand-insert to original board side.

CN3 | CPL 90 deg | Top

USB4105-GF-A

XY: 94.662, 64.765 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

CN4 | CPL 180 deg | Top

USB4105-GF-A

XY: 59.58, 148.583 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

D1 | CPL 90 deg | Top

ESD8351XV2T1G

XY: 73.0, 138.35 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

D2 | CPL 270 deg | Top

WR148SR-W

XY: 30.825, 72.275 mm



Exact purchased part / tape zero pending

cathode pad checked Exact WR148SR-W source or qualified replacement must map its cathode mark to board pad 1. Factory derives tape rotation from supplied part.

D3 | CPL 270 deg | Top

WR148SR-W

XY: 32.75, 72.265 mm



Exact purchased part / tape zero pending

cathode pad checked Exact WR148SR-W source or qualified replacement must map its cathode mark to board pad 1. Factory derives tape rotation from supplied part.

D4 | CPL 270 deg | Top

WR148SR-W

XY: 34.675, 72.275 mm

Exact purchased part / tape zero pending

cathode pad checked Exact WR148SR-W source or qualified replacement must map its cathode mark to board pad 1. Factory derives tape rotation from supplied part.

D5 | CPL 270 deg | Top

SMBJ6.5CA

XY: 83.197, 15.035 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

D6 | CPL 180 deg | Top

WR148SR-W

XY: 79.573, 9.725 mm



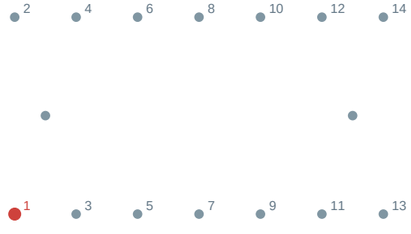
Exact purchased part / tape zero pending

cathode pad checked Exact WR148SR-W source or qualified replacement must map its cathode mark to board pad 1. Factory derives tape rotation from supplied part.

J1 | CPL 0 deg | Top

FTSH-107-01-L-DV-K-A

XY: 78.84, 149.06 mm



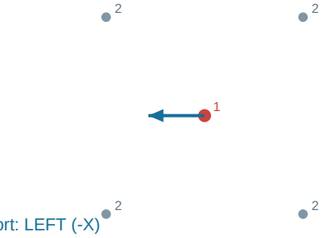
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

J2 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 140.0 mm



Mating port: LEFT (-X)

Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

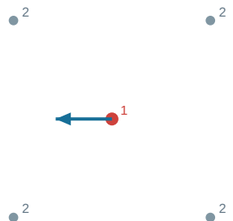
Numbered-pad direction review 13-24

All detail plots are top view, +X right / +Y up. Red marks pin 1, A1 or LED cathode; plots are not to scale.

J3 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 126.0 mm



Mating port: LEFT (-X)

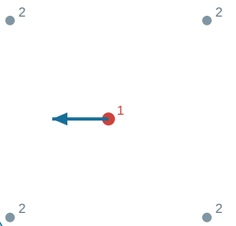
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J4 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 112.0 mm



Mating port: LEFT (-X)

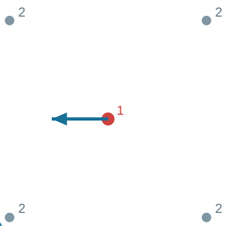
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J5 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 98.0 mm



Mating port: LEFT (-X)

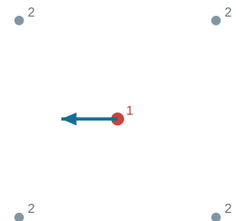
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J6 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 84.0 mm



Mating port: LEFT (-X)

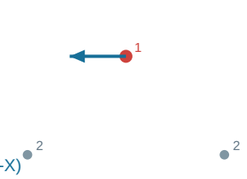
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J7 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 70.0 mm



Mating port: LEFT (-X)

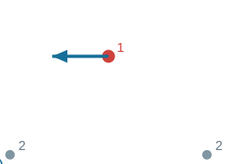
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J8 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 56.0 mm



Mating port: LEFT (-X)

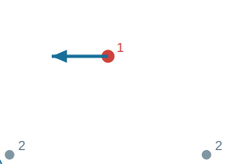
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J9 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 42.0 mm



Mating port: LEFT (-X)

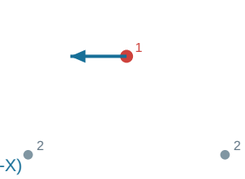
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J10 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 28.0 mm



Mating port: LEFT (-X)

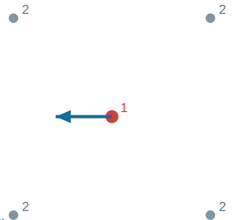
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J11 | CPL 90.0 deg | Top

SMA-KWE901

XY: 4.5, 14.0 mm



Mating port: LEFT (-X)

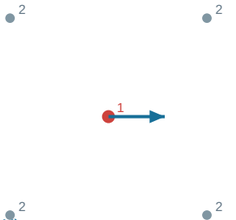
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J12 | CPL 270.0 deg | Top

SMA-KWE901

XY: 95.5, 140.0 mm



Mating port: RIGHT (+X)

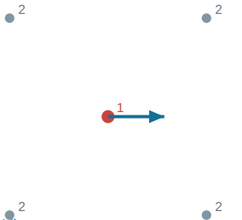
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J13 | CPL 270.0 deg | Top

SMA-KWE901

XY: 95.5, 126.0 mm



Mating port: RIGHT (+X)

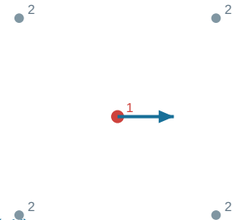
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J14 | CPL 270.0 deg | Top

SMA-KWE901

XY: 95.5, 98.0 mm



Mating port: RIGHT (+X)

Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

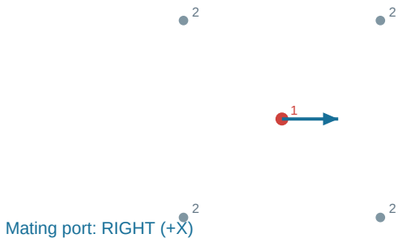
Numbered-pad direction review 25-36

All detail plots are top view, +X right / +Y up. Red marks pin 1, A1 or LED cathode; plots are not to scale.

J15 | CPL 270.0 deg | Top

SMA-KWE901

XY: 95.5, 112.0 mm



Mating port: RIGHT (+X)

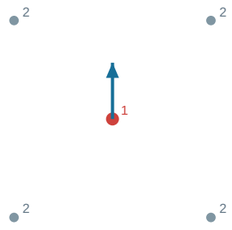
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J16 | CPL 0.0 deg | Top

SMA-KWE901

XY: 40.54, 149.42 mm



Mating port: UP (+Y)

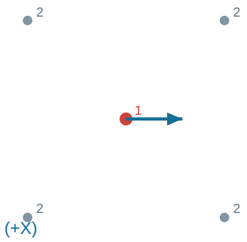
Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J17 | CPL 270.0 deg | Top

SMA-KWE901

XY: 95.5, 84.0 mm



Mating port: RIGHT (+X)

Catalog pin mapping checked

port and signal checked Confirm in exact part placement preview against numbered pad drawing.

J18 | CPL 0 deg | Top

PPTC052LFBN-RC

XY: 42.685, 5.205 mm



Exact purchased part / tape zero pending

manual insertion drawing checked Confirm exact connector outline, pin 1 and mating direction from drawing; hand-insert to original board side.

Q1 | CPL 0 deg | Top

9013M

XY: 71.35, 56.755 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

SW1 | CPL 0 deg | Top

GT-TC089B-H075-L1

XY: 78.75, 139.08 mm



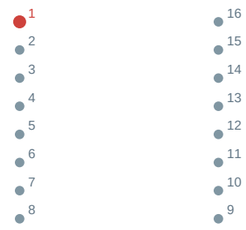
Catalog pin mapping checked

switch net pairs checked Confirm in exact part placement preview against numbered pad drawing.

TR1 | CPL 270 deg | Top

KMS-1102NL

XY: 74.537, 37.525 mm



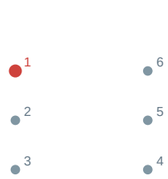
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

TVS1 | CPL 270 deg | Top

USBLC6-4SC6

XY: 62.95, 37.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

TVS2 | CPL 270 deg | Top

USBLC6-2SC6

XY: 87.315, 64.765 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

TVS3 | CPL 0 deg | Top

USBLC6-2SC6

XY: 59.58, 141.81 mm



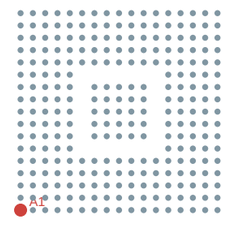
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U1 | CPL 90 deg | Top

STM32H747XIH6

XY: 50.05, 86.725 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U2 | CPL 0 deg | Top

LD1117S18TR

XY: 64.2, 50.475 mm



Catalog pin mapping checked

lead row and tab checked Confirm in exact part placement preview against numbered pad drawing.

Numbered-pad direction review 37-48

All detail plots are top view, +X right / +Y up. Red marks pin 1, A1 or LED cathode; plots are not to scale.

U3 | CPL 270 deg | Top

ST1L05CPU33R

XY: 70.862, 15.535 mm



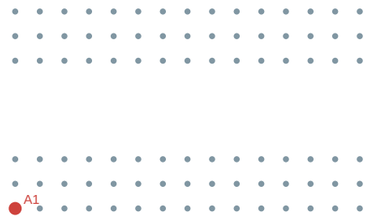
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U4 | CPL 0 deg | Top

IS42S32160F-6BLI

XY: 46.6, 114.15 mm



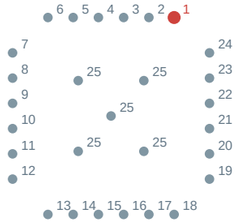
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U5 | CPL 270 deg | Top

LAN8742A-CZ-TR

XY: 52.55, 37.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U6 | CPL 90 deg | Top

STS5PF30L

XY: 64.3, 57.985 mm



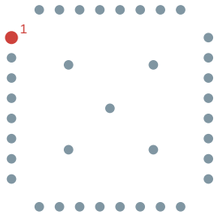
Exact purchased part / tape zero pending

so8 pin order checked Resolve obsolete part sourcing or approve a separately reviewed substitute; verify purchased tape pin-1 orientation against drawing before assembly.

U7 | CPL 270 deg | Top

USB3320C-EZK

XY: 76.625, 66.65 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U8 | CPL 90 deg | Top

CH340C

XY: 59.63, 131.335 mm



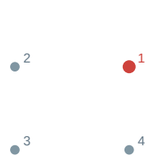
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

X1 | CPL 180 deg | Top

ECS-3225MV-250-CN-TR

XY: 54.65, 72.425 mm



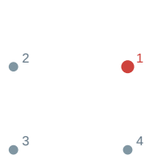
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

X2 | CPL 180 deg | Top

ECS-3225MV-240-CN-TR

XY: 82.312, 76.775 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

Y1 | CPL 0 deg | Top

NX3215SA-32.768KHZ-EXS00A-MU00466

XY: 39.626, 72.185 mm



Catalog pin mapping checked

nonpolar axis checked Check custom hand-solder land coverage against NDK NX3215SA dimensions; do not change orientation by 90 degrees.

U10 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 138.0 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U11 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 124.0 mm



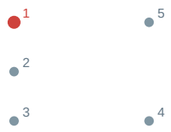
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U12 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 110.0 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

Numbered-pad direction review 49-60

All detail plots are top view, +X right / +Y up. Red marks pin 1, A1 or LED cathode; plots are not to scale.

U13 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 96.0 mm



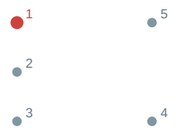
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U14 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 82.0 mm



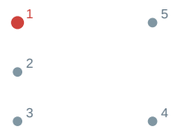
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U15 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 68.0 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U16 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 54.0 mm



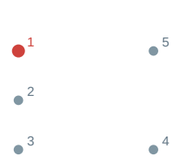
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U17 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 40.0 mm



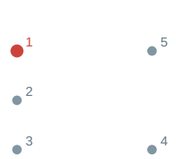
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U18 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 26.0 mm



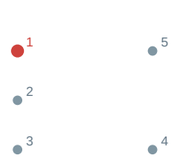
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U19 | CPL 180 deg | Top

OPA320AIDBVR

XY: 18.8, 12.0 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

D20 | CPL 180 deg | Top

BAT54WS-E3-08

XY: 34.2, 33.2 mm



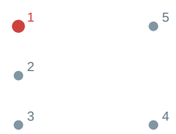
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U20 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 136.5 mm



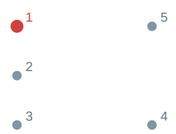
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U21 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 122.5 mm



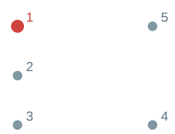
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U22 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 108.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U23 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 94.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

Numbered-pad direction review 61-70

All detail plots are top view, +X right / +Y up. Red marks pin 1, A1 or LED cathode; plots are not to scale.

U24 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 80.5 mm



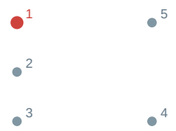
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U25 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 66.5 mm



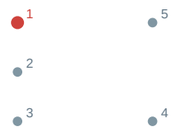
Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U26 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 52.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U27 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 38.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U28 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 24.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U29 | CPL 270 deg | Top

OPA189IDBVR

XY: 11.2, 10.5 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U30 | CPL 180 deg | Top

TPS60403DBVR

XY: 31.0, 29.8 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U31 | CPL 180 deg | Top

TPS72301DBVR

XY: 38.0, 29.8 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U32 | CPL 270 deg | Top

OPA189IDBVR

XY: 25.7, 70.9 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

U9 | CPL 0 deg | Top

REF5025AID

XY: 26.1, 76.9 mm



Catalog pin mapping checked

numbered pads checked Confirm in exact part placement preview against numbered pad drawing.

Factory review and release conditions

Local verification is complete. Sourcing allocation, purchased-part tape zero and production release are pending.

Files and exclusions

Use BOM_JLCPCB.csv and CPL_JLCPCB_REVIEW.csv together. 69 BOM groups total 412 parts/board. Exclude C33, R61, R64, R98, R125, R126 (DNP) and 19 bare test pads. Native CPL has 437 rows and must not be uploaded as the assembly list.

Legacy procurement

139 original parts have no verified JLC assembly SKU. Match exact manufacturer part numbers through Manual Match or approved Global Sourcing. U6 STS5PF30L is obsolete per ST. Do not substitute or omit it without a separately reviewed decision.

Seven purchased-part direction checks

CN2, D2, D3, D4, D6, J18 and U6: numbered-pad / cathode direction has been checked locally. CPL angles use a documented drawing convention; exact supplier tape zero is not asserted. Match purchased parts to these pad drawings before assembly.

Special lands

Y1 NX3215SA: nonpolar X-axis; extended hand-solder lands need contact-coverage DFM. U2 SOT-223 and U6 SO-8 also retain extended native lands. CN2/J18 are manual-insertion references; confirm pin 1, mating direction, insertion side and mechanical fit.

SMA orientation

J2-J11: left port, CPL 90 deg. J12-J15 and J17: right port, CPL 270 deg. J16: upper port, CPL 0 deg. Derived from the identical supplier 3D model UUID and its 180 deg canonical orientation, not an arbitrary visual zero.

CAD checks

Strict PCB DRC: 0. Native ERC: fatal 0 / error 0, supplier-property warning 5 groups retained. The 132 selected parts use original custom symbols and footprints. All copper, mask, paste, silk, outline and drill manufacturing geometry is preserved.

Production settings

6 layers, finished 100 x 154 mm, 1.6 mm draft, ENIG, Standard PCBA both sides. Native min via hole 0.2 mm / diameter 0.31 mm. Confirm original stackup, layer order, PCB production files and parts placement. Quantity remains unconfirmed; 5+5 website quote is a draft.

Source references

JLCPCB: Pick & Place File for PCB Assembly; PCB Assembly FAQs Part 2. ST: STS5PF30L product page and datasheet. NDK: NX3215SA catalog. Source URLs and exact geometry evidence are included in ROTATION_REVIEW.json and CAM_AUDIT.json.